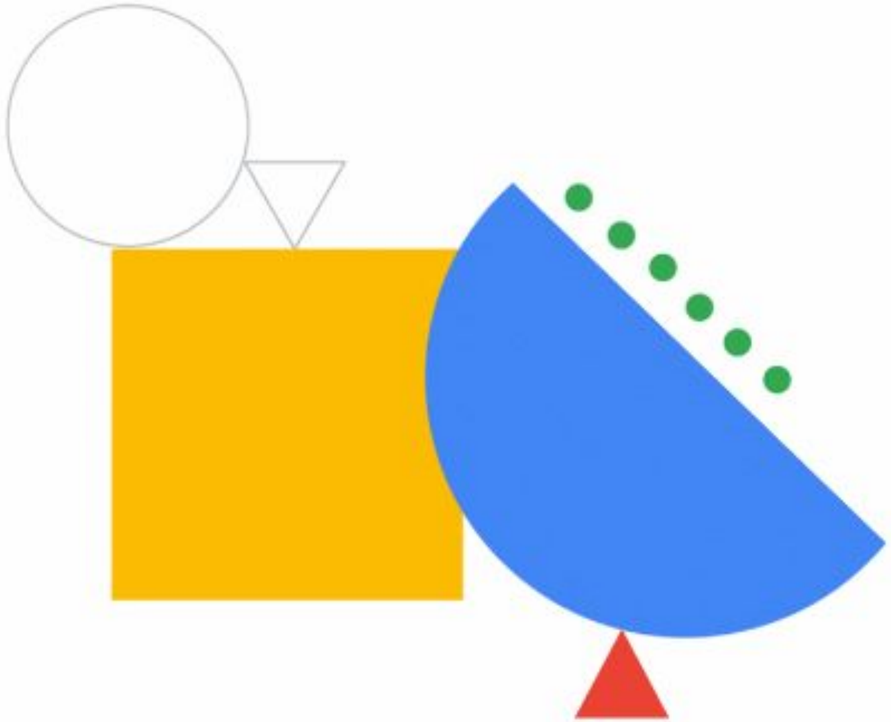
Google Cloud

Monitoring Critical Systems



Proprietary + Confidential

Objectives

01 Use Cloud Monitoring to view metrics for multiple cloud projects

02 Explain the different types of dashboards and charts that can be built

03 Create an uptime check

04 Explain the purpose of using MQL for monitoring



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In this section, you explore



- Monitoring overview
- Cloud Monitoring architecture patterns
- Monitoring multiple projects
- Data model and dashboards
- Query metrics
- Uptime checks

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Monitoring is the foundation of product reliability



Cloud Monitoring

Collecting, processing, aggregating, and displaying **real-time quantitative data about a system**.

- Reveals what needs urgent attention
- Shows trends in application usage patterns
- Helps improve an application experience

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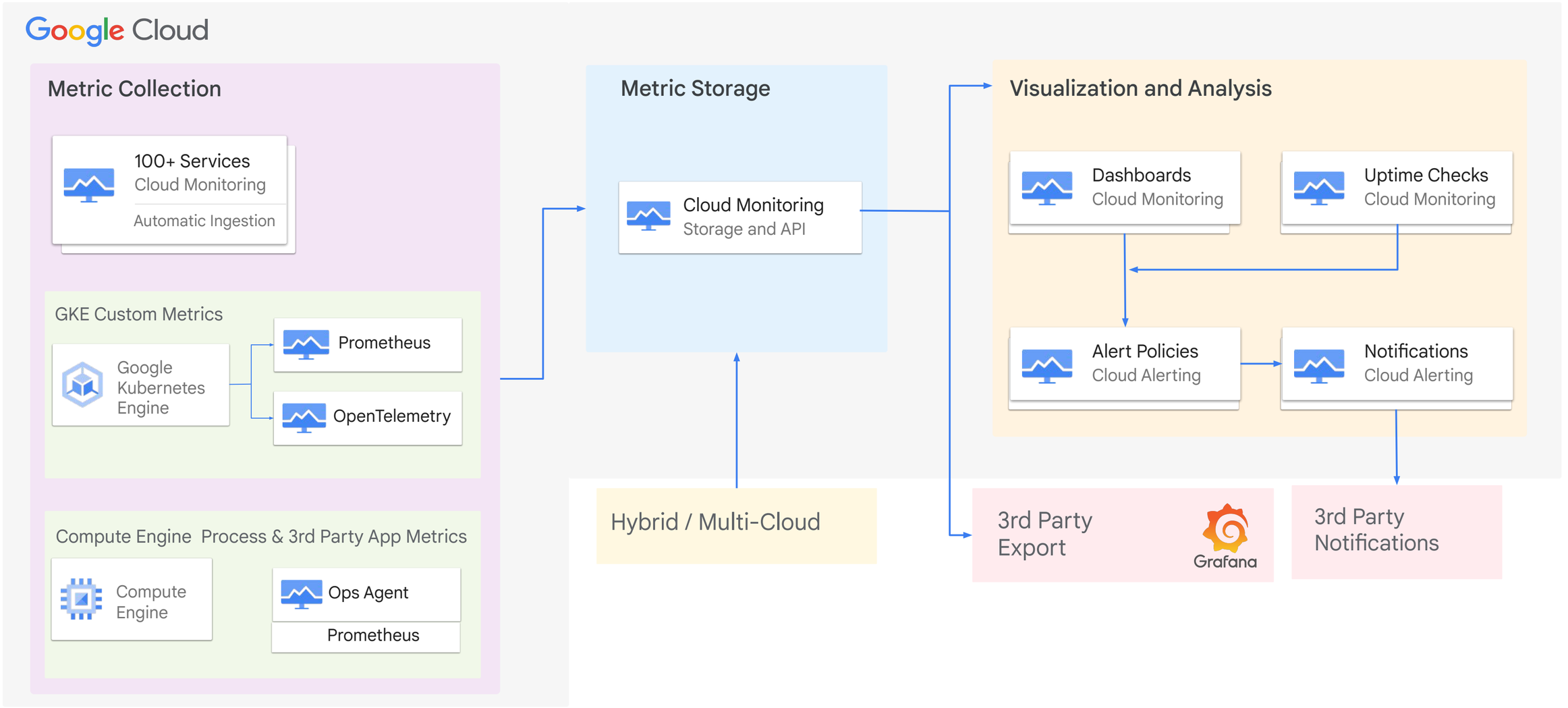
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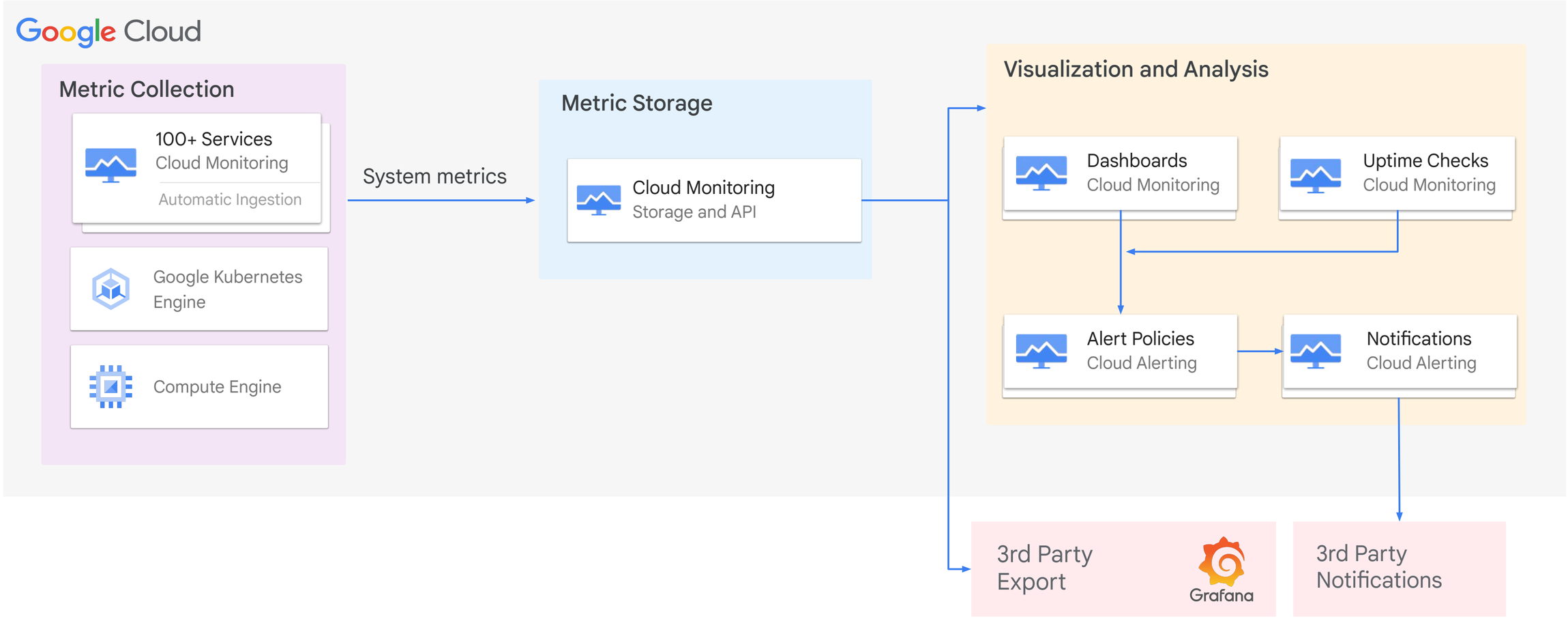
Cloud Monitoring architecture

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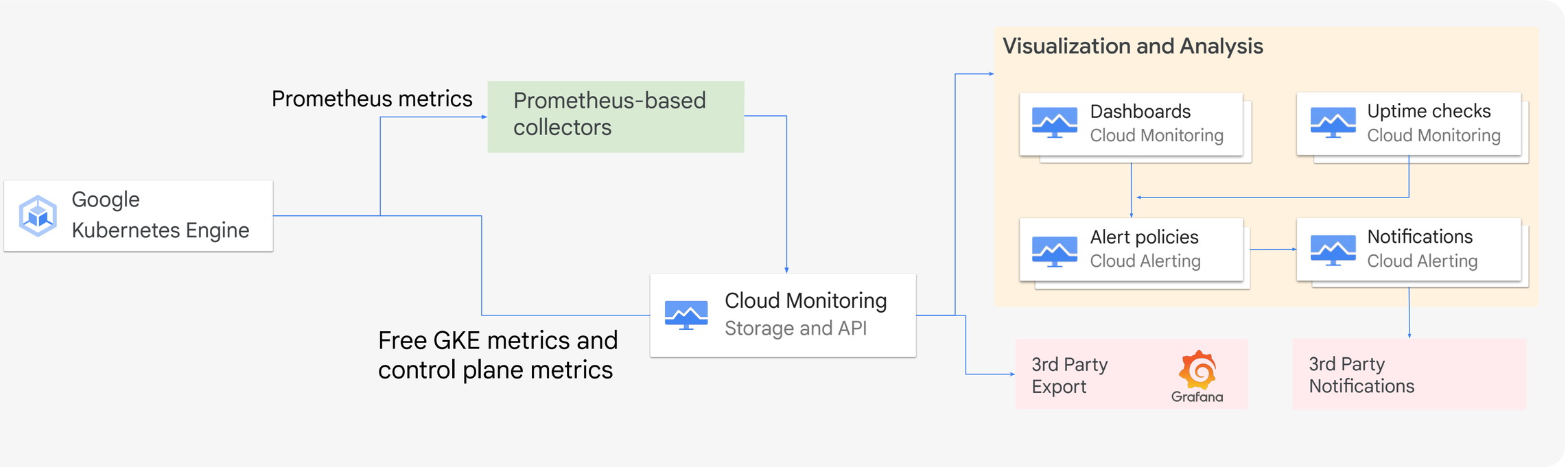
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Platform monitoring



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Application monitoring - GKE

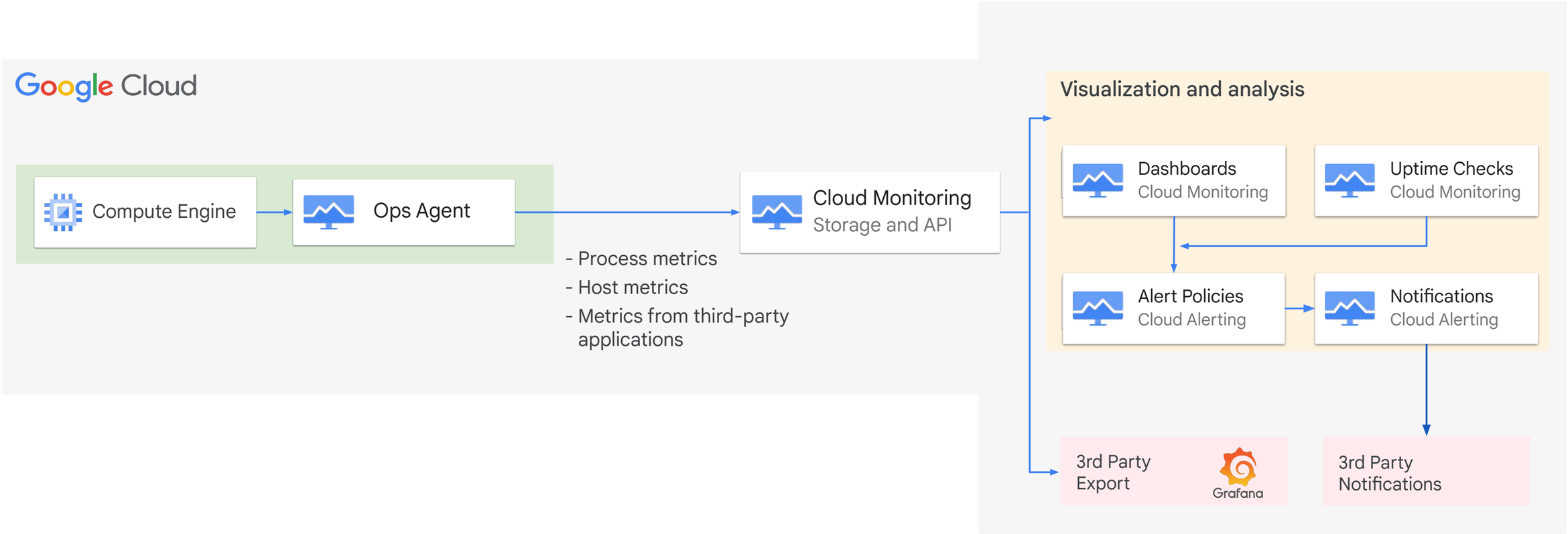


GKE includes integration with:

- Cloud Logging and Cloud Monitoring
- Google Cloud Managed Service for Prometheus

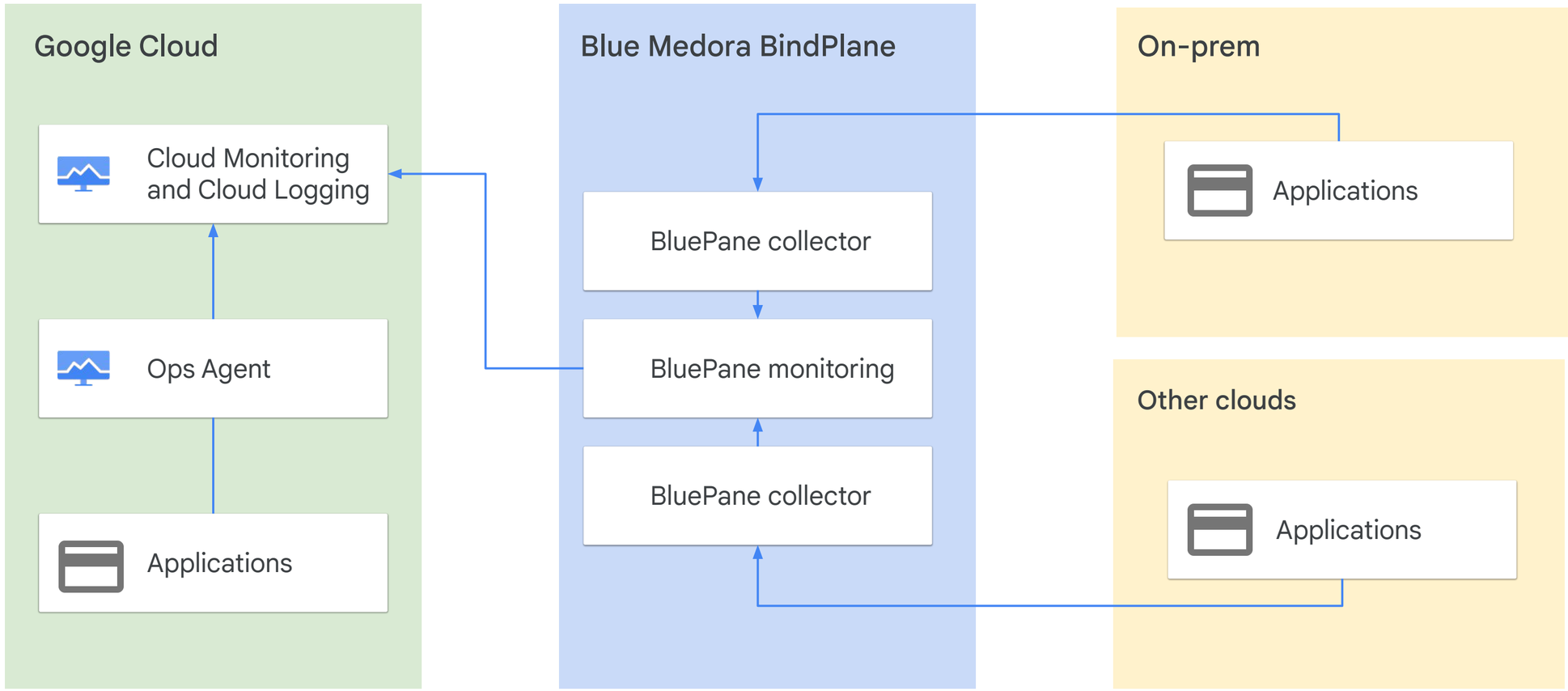
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Application monitoring - Compute Engine



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Hybrid monitoring and logging



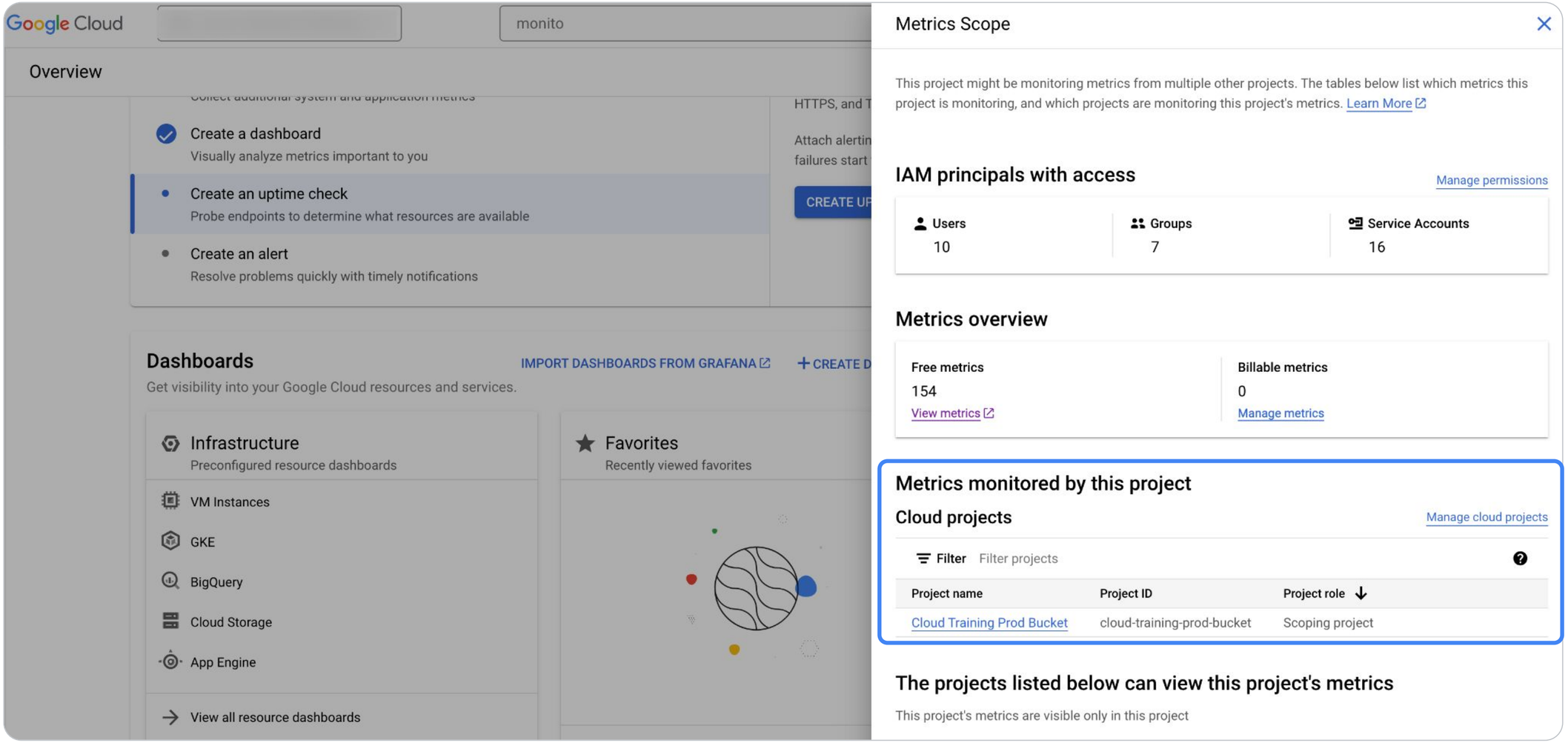
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In this section, you explore

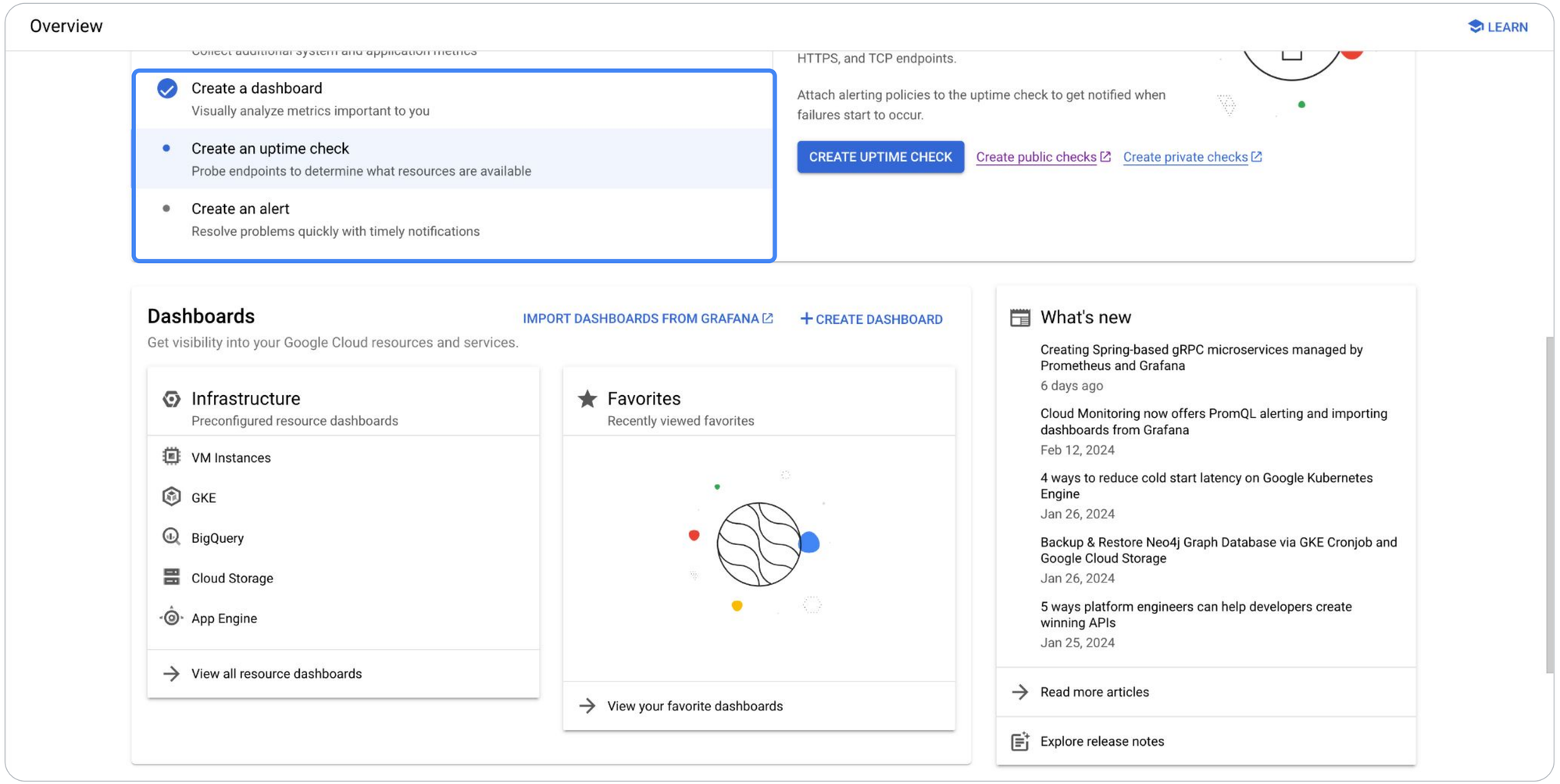


- Monitoring overview
- Cloud Monitoring architecture patterns
- **Monitoring multiple projects**
- Data model and dashboards
- Query metrics
- Uptime checks

The default metrics scope contains the current project

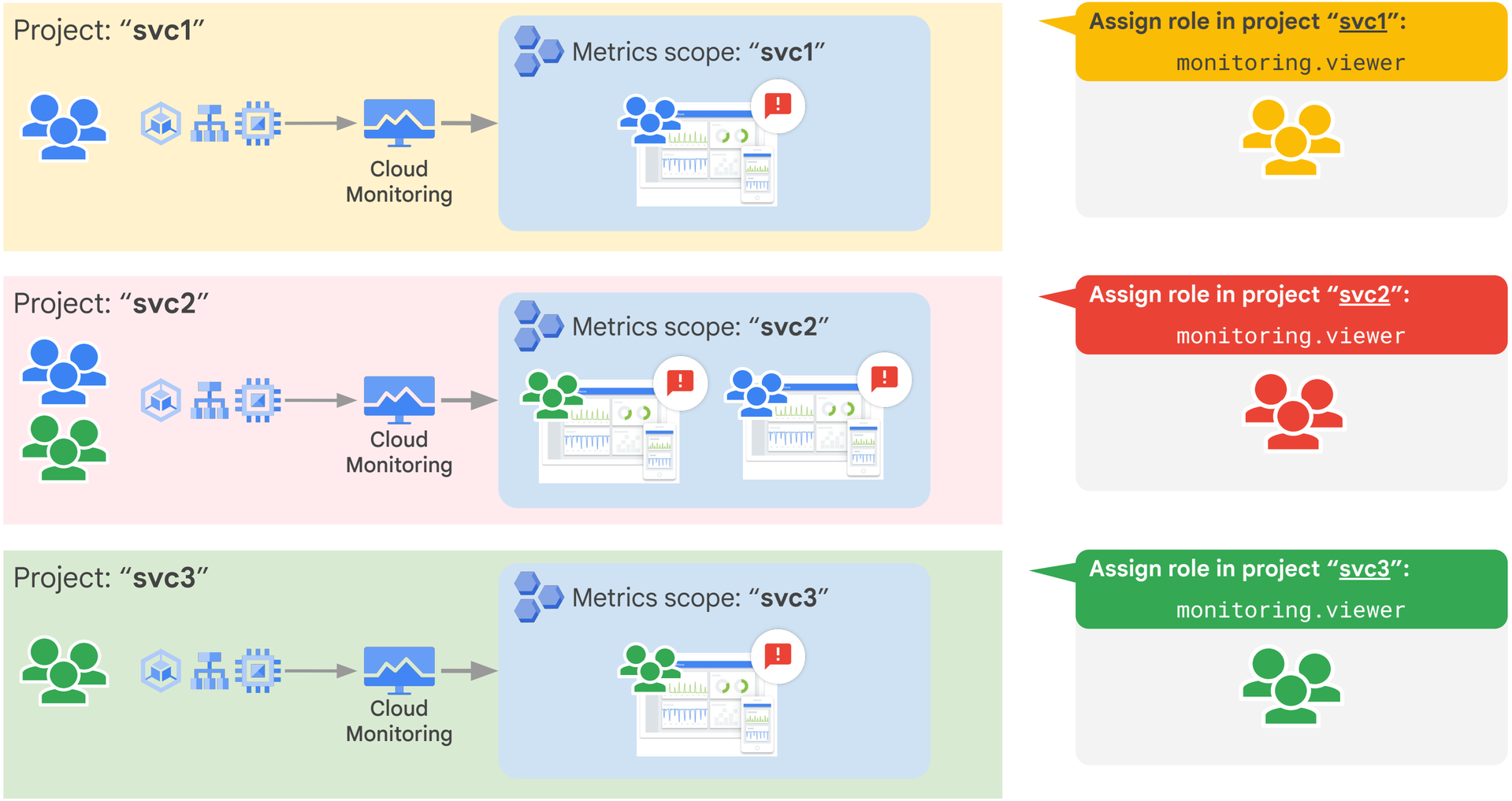


The default metrics scope contains the current project



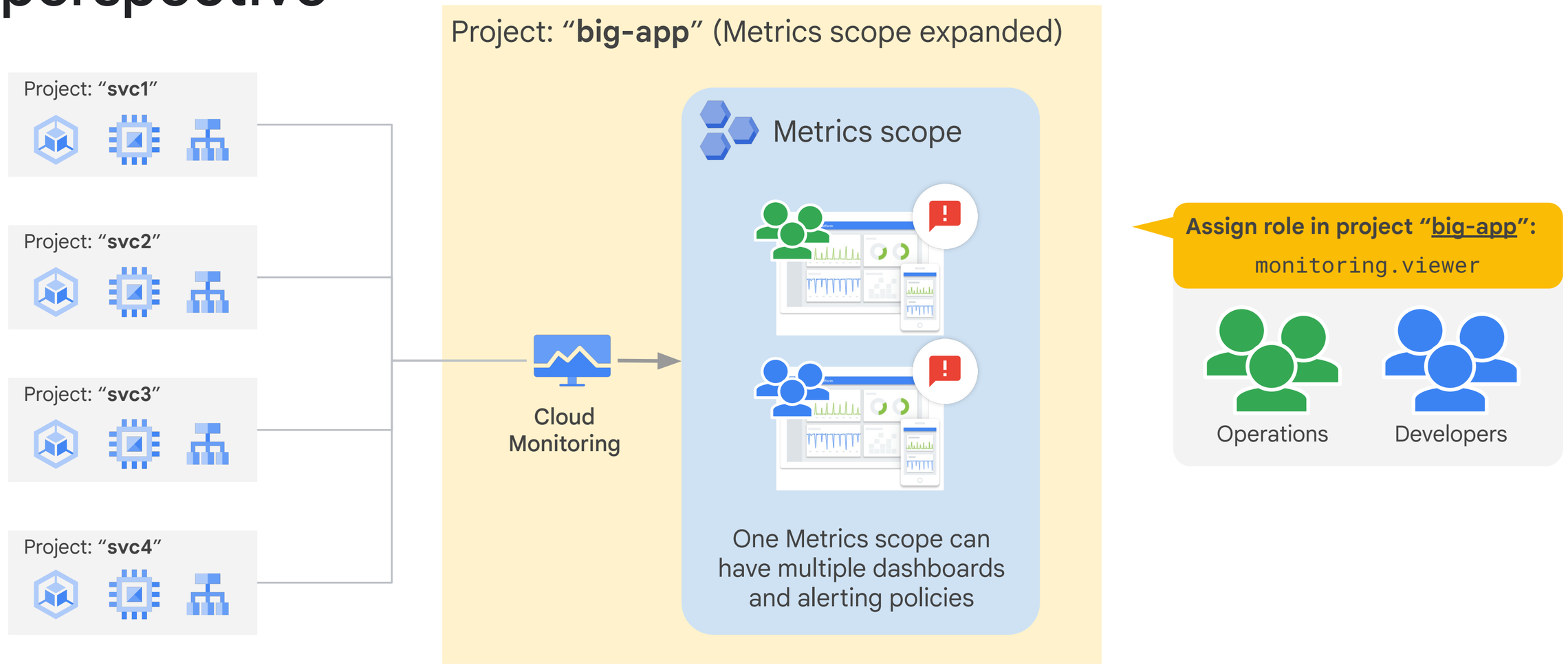
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The default metrics scope contains the current project



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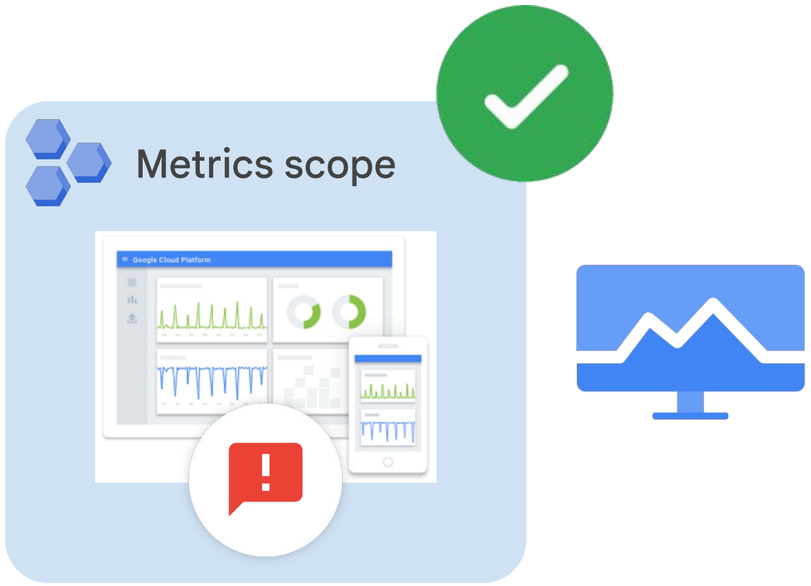
Optionally, create monitoring projects for a wider perspective



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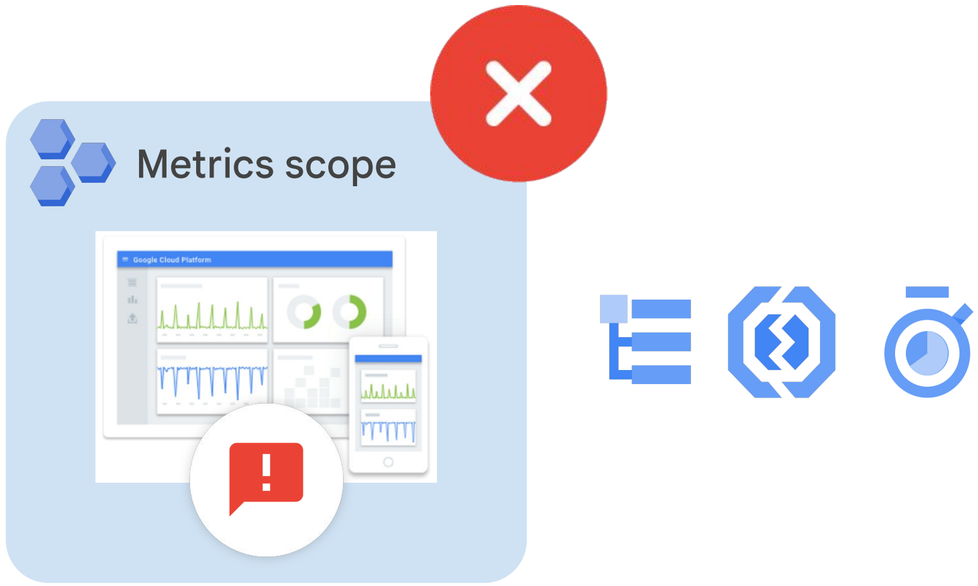
Remember: Metrics scopes only affect Cloud Monitoring

Only the Cloud Monitoring system relies upon metrics scopes



The other tools in this course:

- Are configured on a per-project basis.
- Have their own IAM roles



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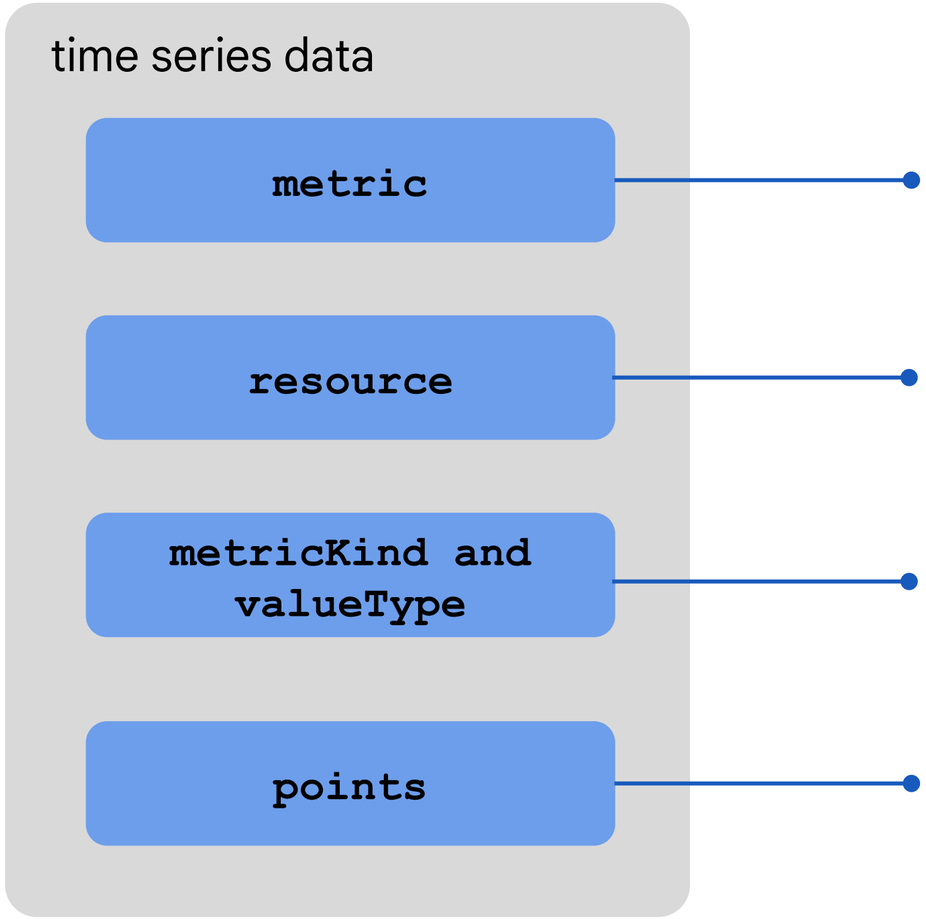
In this section, you explore



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Cloud Monitoring data model



metric field consists of metric label and metric type that describes the metric

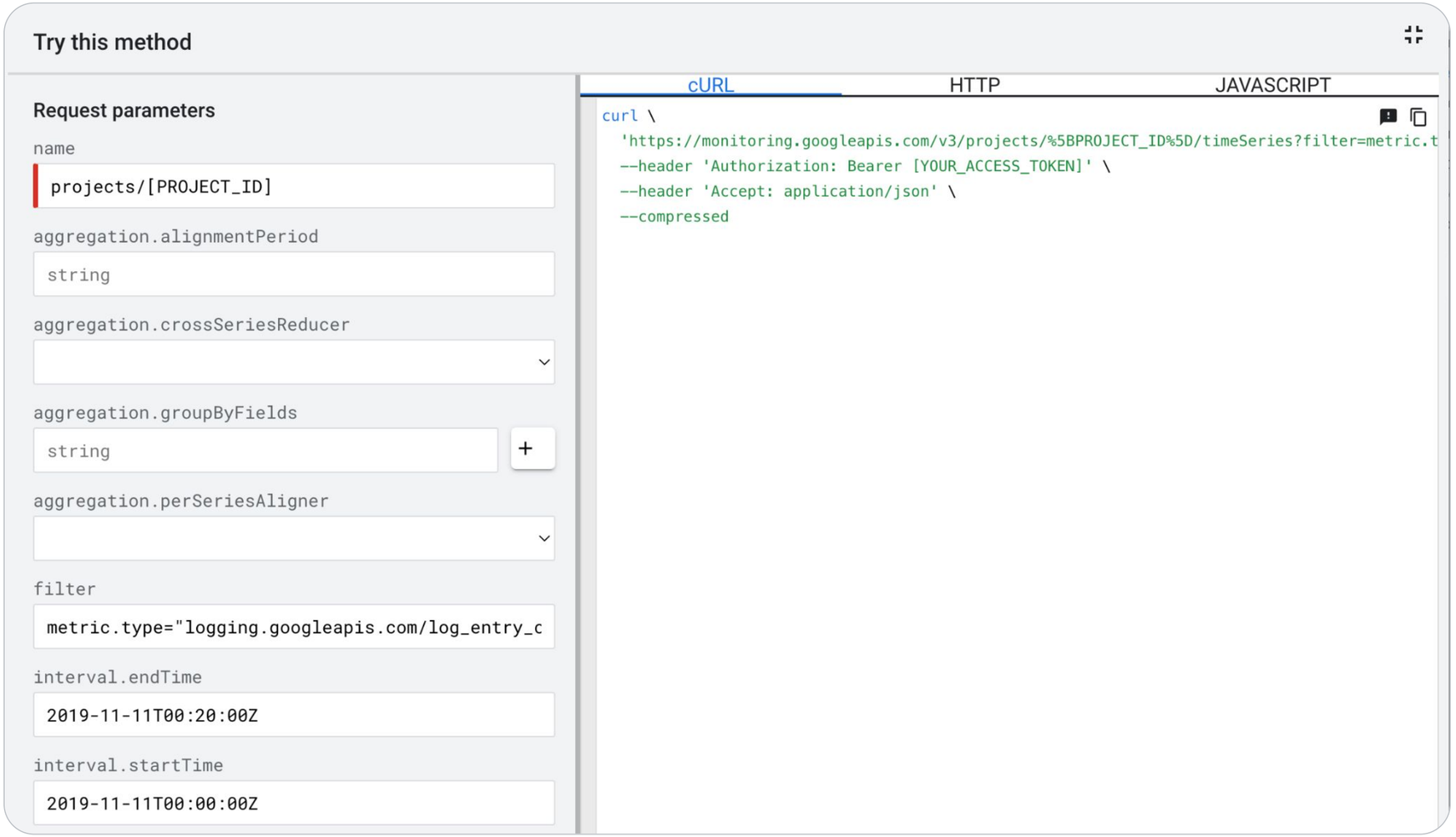
resource field consists of resource-label and the resource information from which the metrics are collected

metricKind and valueType tells you how to interpret the values

points are an array of timestamped values that tells you what the values of the metrics are

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Cloud Monitoring metric type



A sample time series

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```
"timeSeries": [
 {
   "metric": {
      "labels": {
         "severity": "INFO",
         "log": "compute.googleapis.com/activity_log"
       },
       "type": "logging.googleapis.com/log_entry_count"
   },
   "resource": {
      "type": "gce_instance",
      "labels": {
         "instance_id": "5106847938295940291",
         "zone": "us-central1-a",
         "project_id": "a-Google Cloud-project"
       }
     },
      "metricKind": "DELTA",
      "valueType": "INT64",
      "points": [
         {
          "interval": {
            "startTime": "2019-12-20T20:25:38Z",
            "endTime": "2019-12-20T20:26:38Z"
          },
          "value": {
            "int64Value": "20"
```


Metric collected is a set of activity logs of the type **log_entry_count** with a severity level INFO

A Compute Engine instance with the specified instance_id, zone and project_id.

The metric kind is of the type DELTA and value type integer.

Points are the actual array of time stamp and value of the metric.

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To monitor your Google Cloud systems:

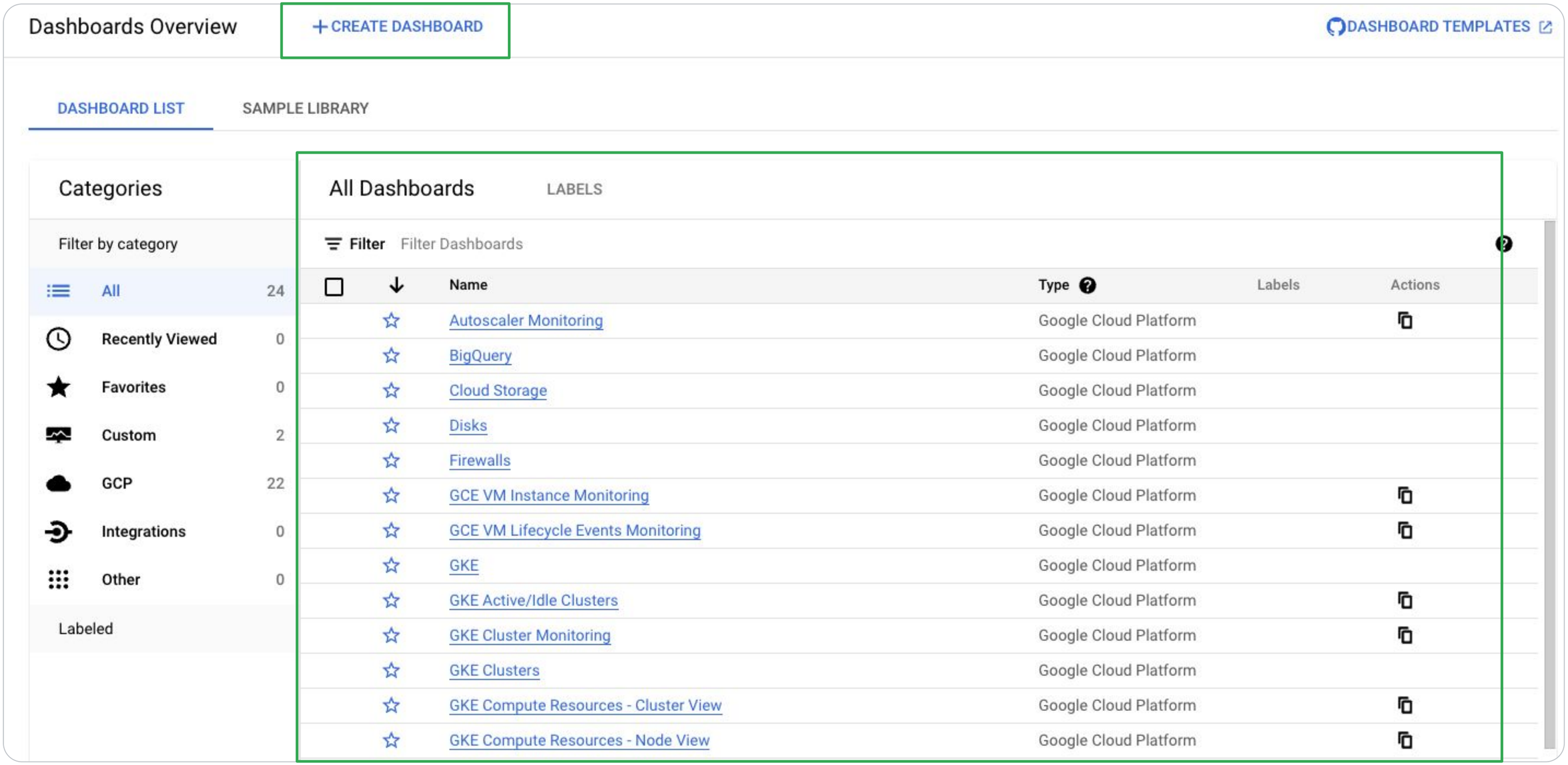
1. Identify the Google Cloud monitoring resources you want to monitor
2. Next, check the **Monitoring Dashboards** for auto-created dashboards
3. When you can’t find what you need, use the **Monitoring Metrics Explorer**



You can monitor any of the more than 1500 metrics, custom metrics and can even build custom dashboards

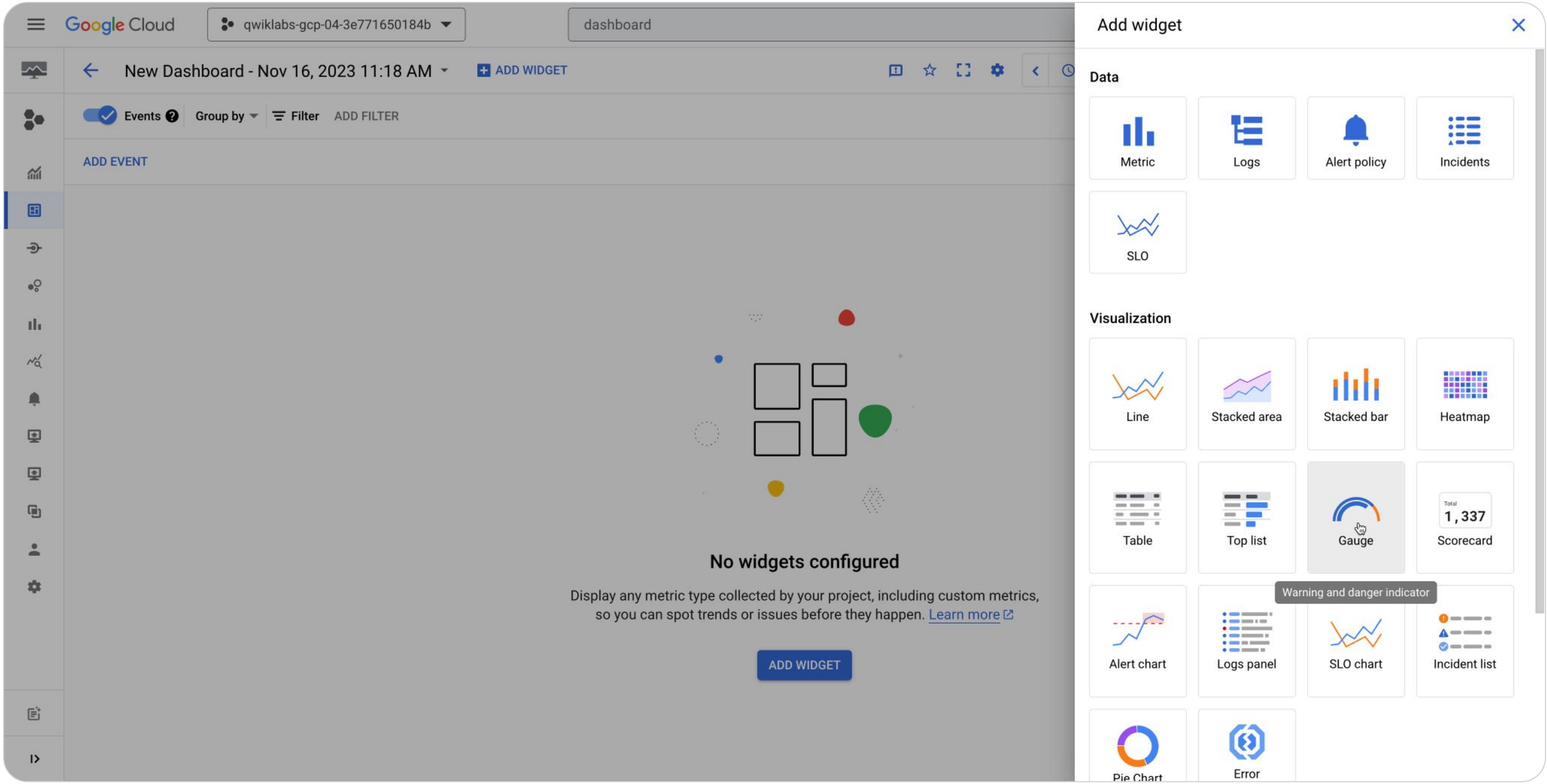
Predefined and custom dashboards

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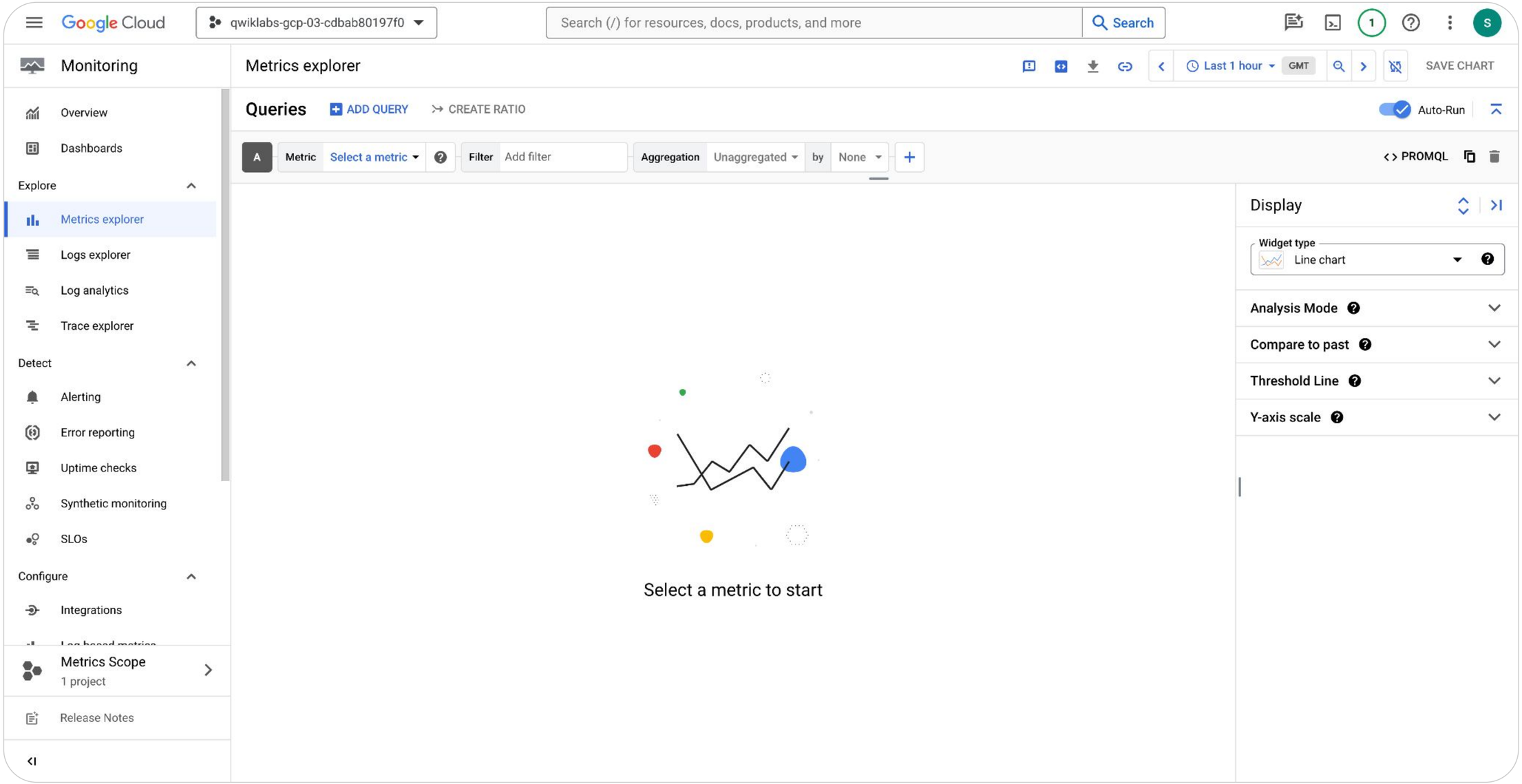
Dashboard Builder

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Metrics Explorer

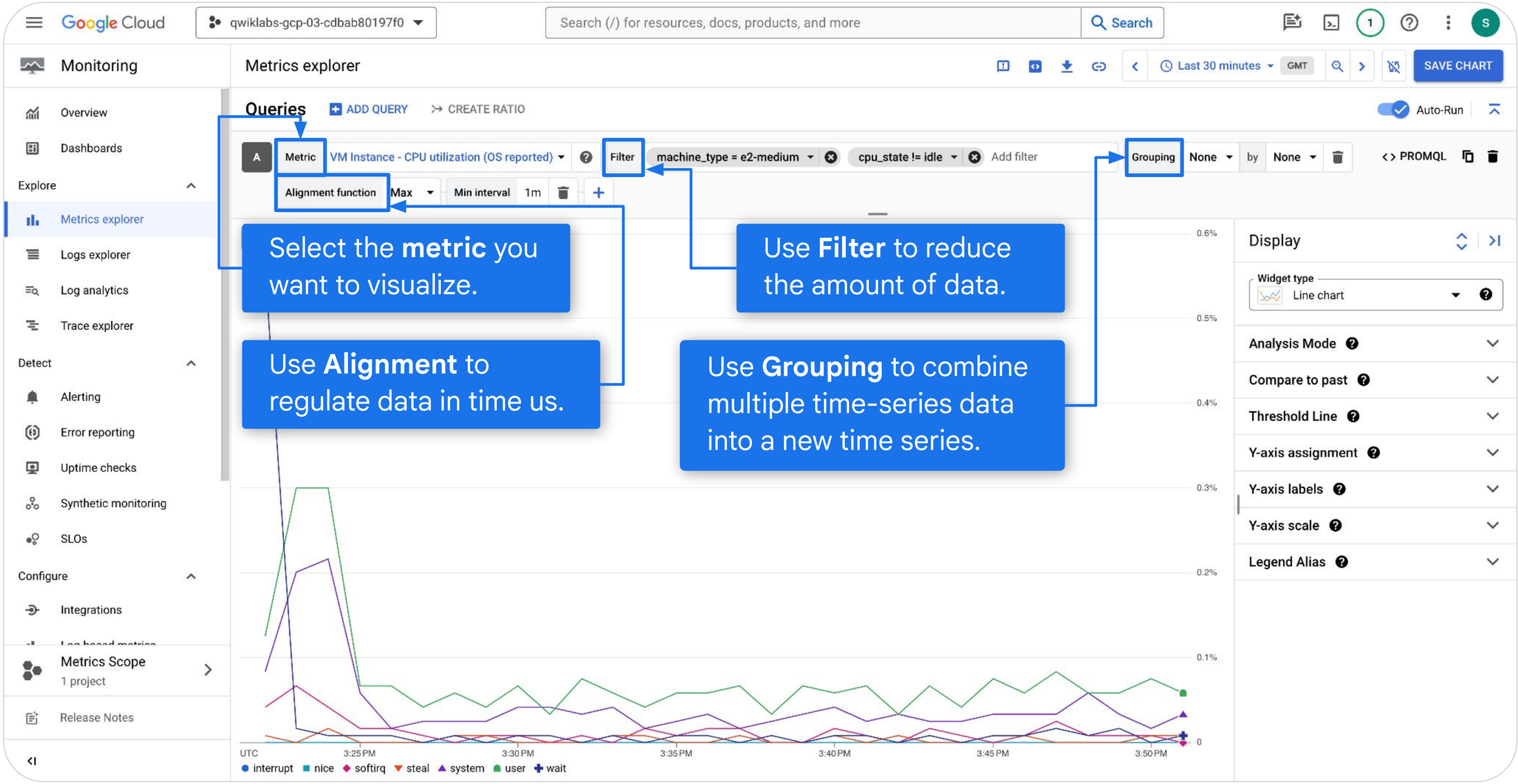
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Google Cloud

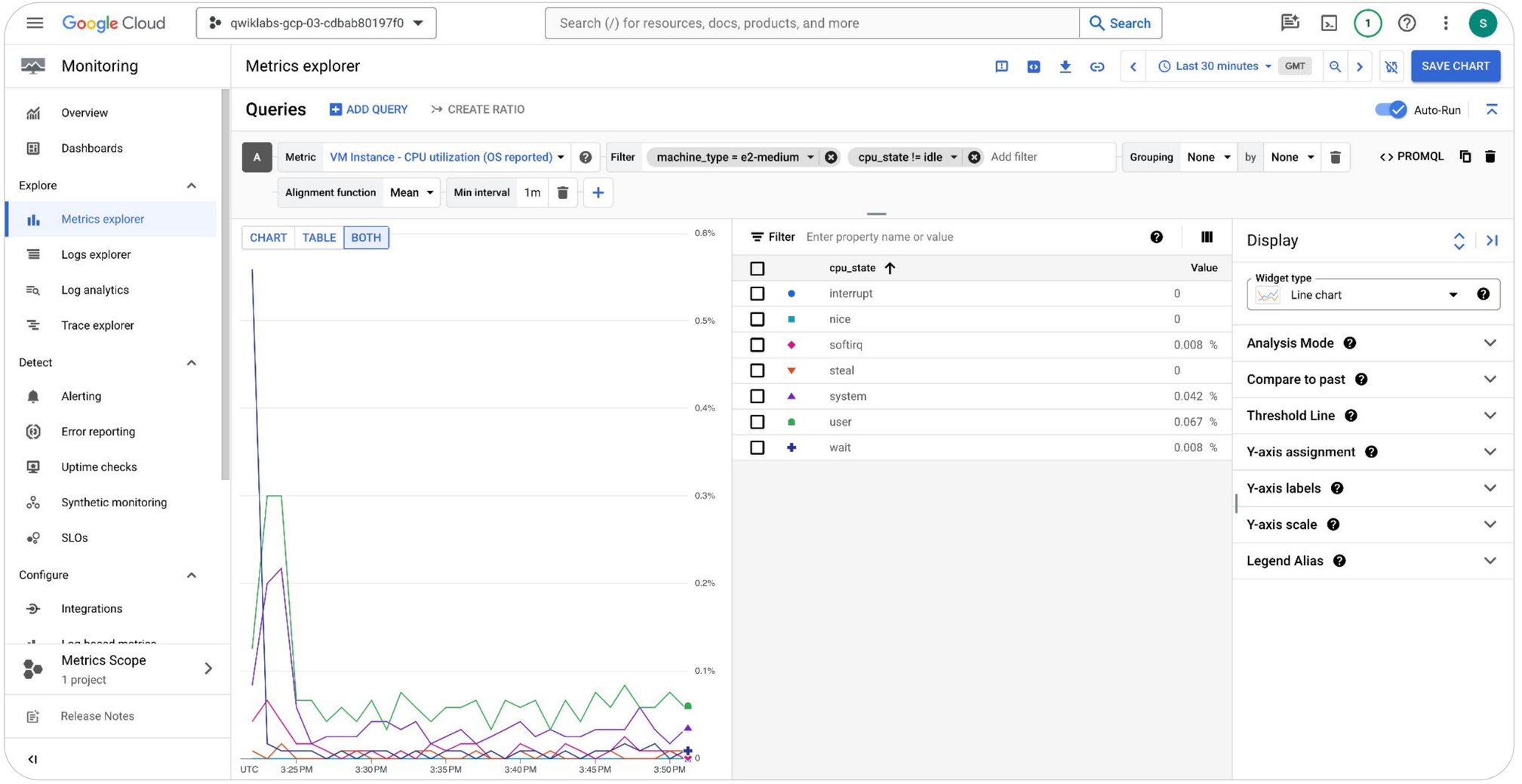
Metrics Explorer

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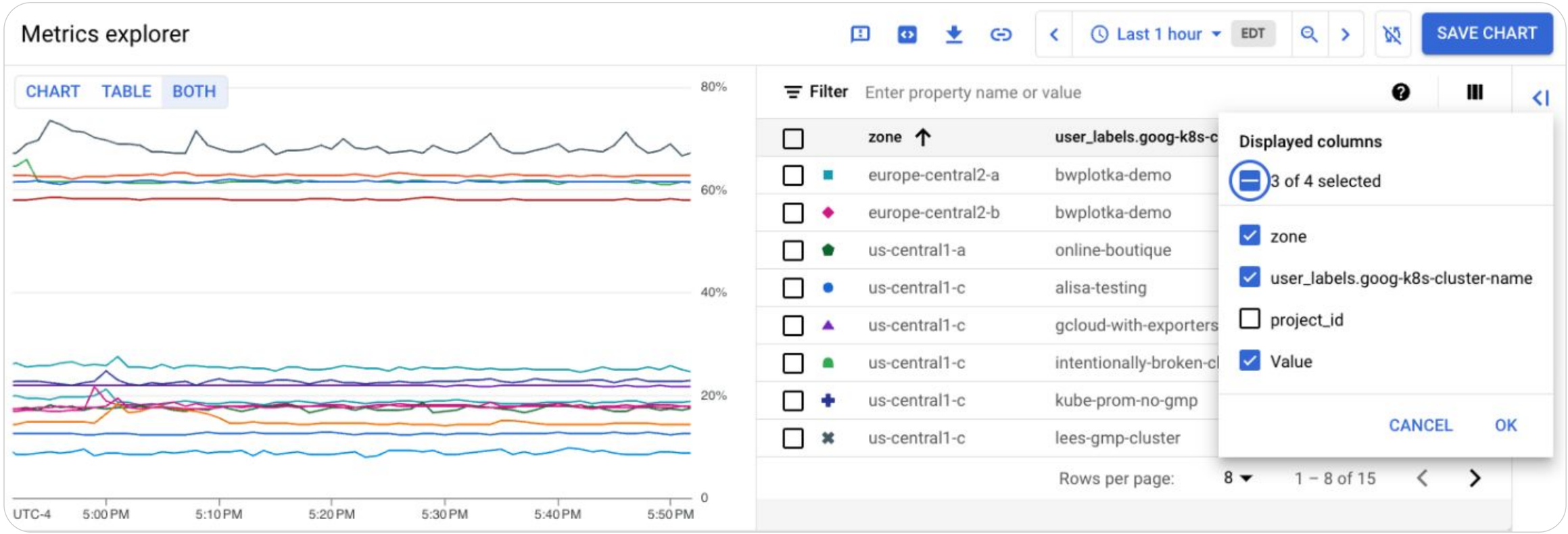
Alignment

Break data into regular time buckets



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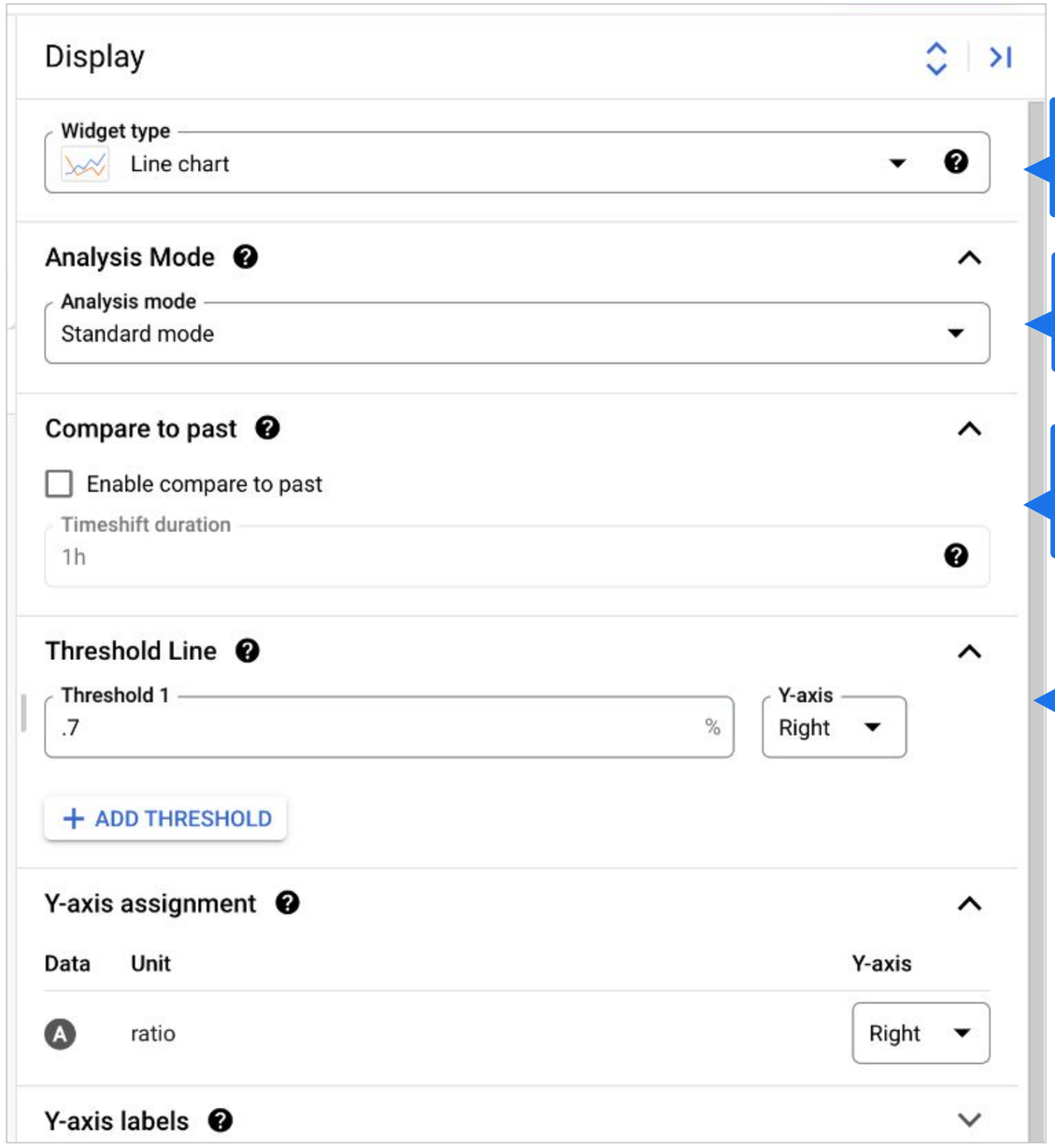
Legend configuration



Google Cloud

Display options

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Select the type of graph from **Widget type**

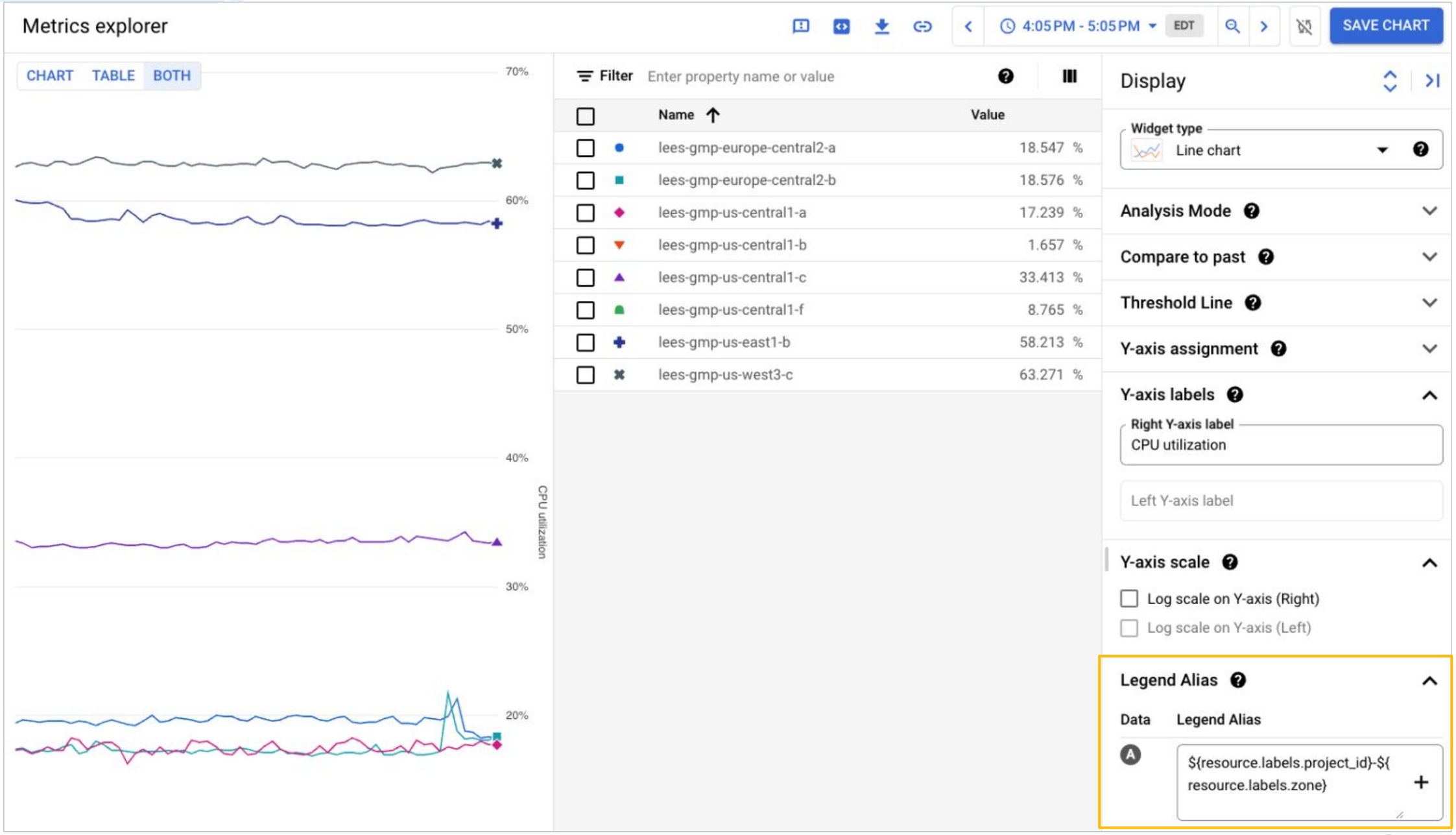
Specify how data is presented in **Analysis mode**

Modify the legend to include a second “values” column by selecting **Enable compare to past**

Create a horizontal line from a point on Y-axis by choosing **a threshold** value.

Legend Alias

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Google Cloud

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In this section, you explore



- Monitoring overview
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- **Query metrics**
- Uptime checks

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Query metrics by using MQL and PromQL

MQL	• Query Cloud Monitoring time-series data by using the Text based interface • Manipulate, retrieve and perform complex operation time-series data
PromQL	• Query metrics from Google Cloud Managed Service for Prometheus • Query System metrics from GKE and Compute Engine • Integrate with Grafana to chart metrics

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Why use MQL?

- Create ratio-based charts and alerts
- Perform time-shift analysis
- Apply mathematical, logical, table operations, and other functions to metrics
- Fetch, join, and aggregate over multiple metrics
- Select by arbitrary, rather than predefined, percentile values
- Create new labels to aggregate data by, using arbitrary string manipulations including regular expressions

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MQL example to analyze error rate

Environment	A distributed web service that runs on **Compute Engine VM** instances and uses **Cloud Load Balancing**
Visualize	A chart that displays requests that return **HTTP 500 responses : total number of requests**.

```
fetch https_lb_rule::loadbalancing.googleapis.com/https/request_count
| group_by [matched_url_path_rule],
    sum(if(response_code_class = 500, val(), 0)) / sum(val())
```

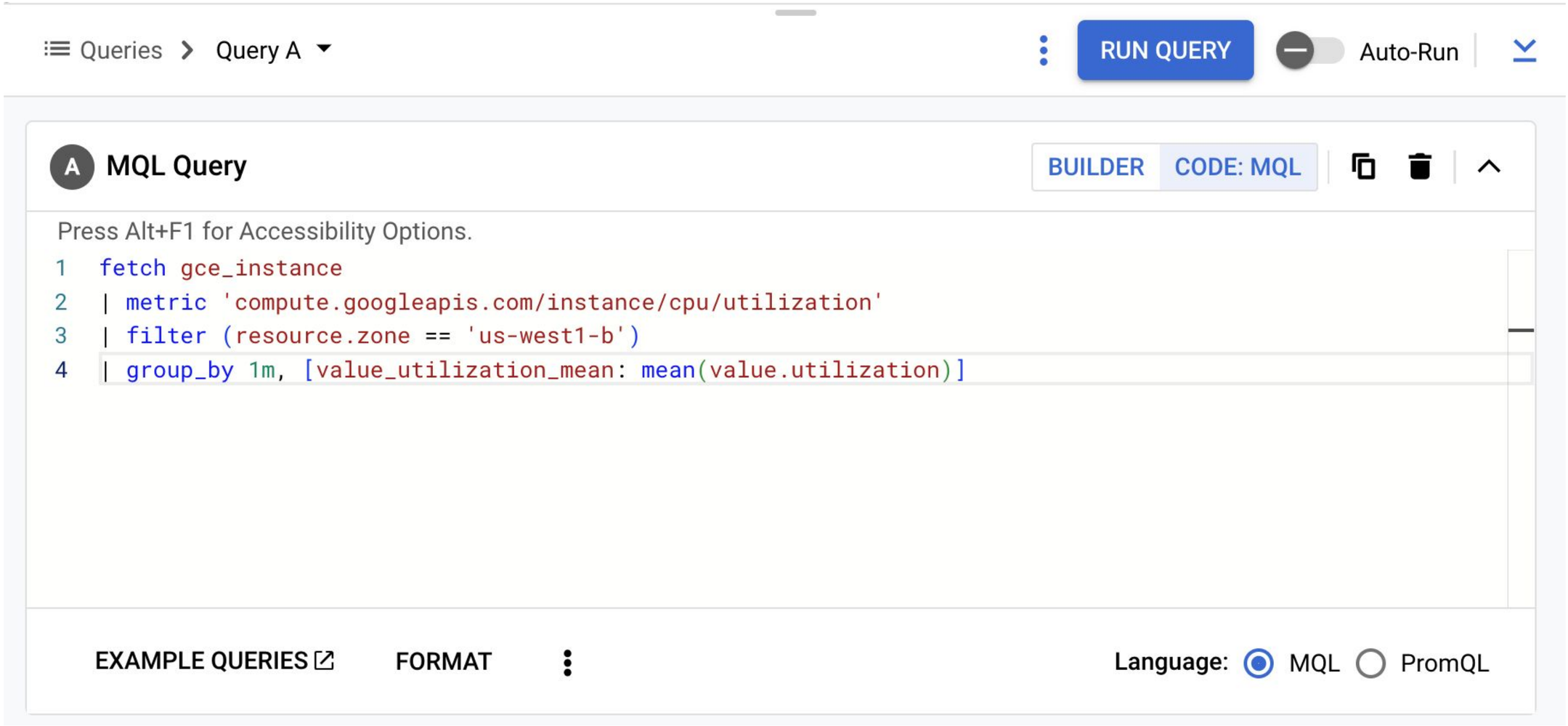
It counts the **number of http 500 responses**

It counts **zero for other http response code**

This is the **total** number of **requests**

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Accessing MQL and PromQL interface



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In this section, you explore

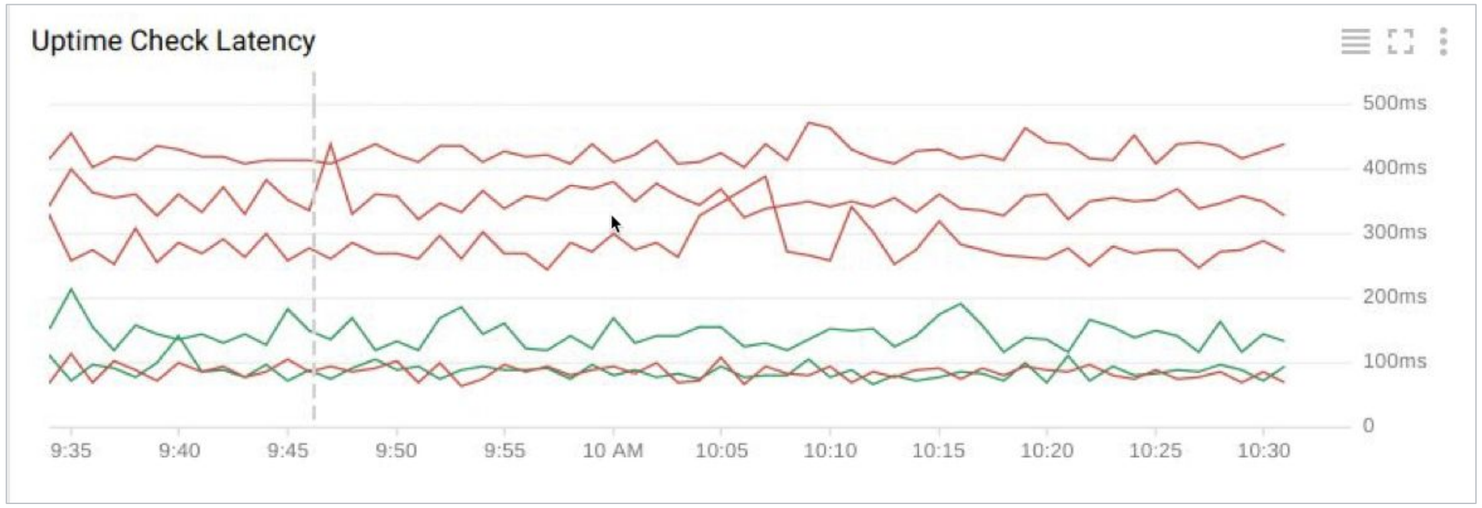


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- **Uptime checks**

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Uptime checks test public service availability

CHECKS	VIRGINIA	OREGON	IOWA	BELGIUM	SINGAPORE	SAO PAULO	POLICIES
Instance 1	✓	✓	✓	✓	✓	✓	
Instance 2	✓	✓	✓	✓	✓	✓	
Instance 3	✓	✓	✓	✓	✓	✓	



Uptime
100.000%

Outages
0 minutes

Location Results All locations passed

Check config

Check Type	HTTP
Resource	summer01
Path	/
Check Every	1 minute
Port	80
Locations	Global
Timeout	10 seconds

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Creating an uptime check

1

Select the resource to be monitored.

Protocol: HTTP

Resource Type: URL

Hostname*: http:// example.com

Path

URL: http://example.com

Check Frequency: 1 minute

Regions
- Global
- Asia Pacific
- Europe
- South America
- United States Iowa
- United States Oregon
- United States Virginia

ICMP Pings: 0 pings

General

Request Method: GET

Host Header

Port: 80

Custom Headers

Encrypt custom headers

Header | Value (optional)

+ ADD CUSTOM HEADER

Authentication

Username | Password

2

Specify data and how that data is to be compared to the actual response data.

Response Timeout *: 10 seconds

Content matching is disabled

Response Content Match Type: Contains

Response Content

Log check failures

Acceptable HTTP Response Code
- Response Code Classes
- Response Codes
- Allow any Response Code

- 1XX
- 2XX
- 3XX
- 4XX
- 5XX

3

Define Uptime Check Alert Condition.

Create an alert

Name *: Uptime failure

Duration: 1 minute

Notifications

When the uptime check fails for the selected duration, you will be notified via these channels.
Learn more

Notification Channels

4

Enter a name for the uptime check.

Title *

User Labels

Key | Value (optional)

+ ADD USER LABEL

HIDE USER LABELS

Test uptime check (optional)

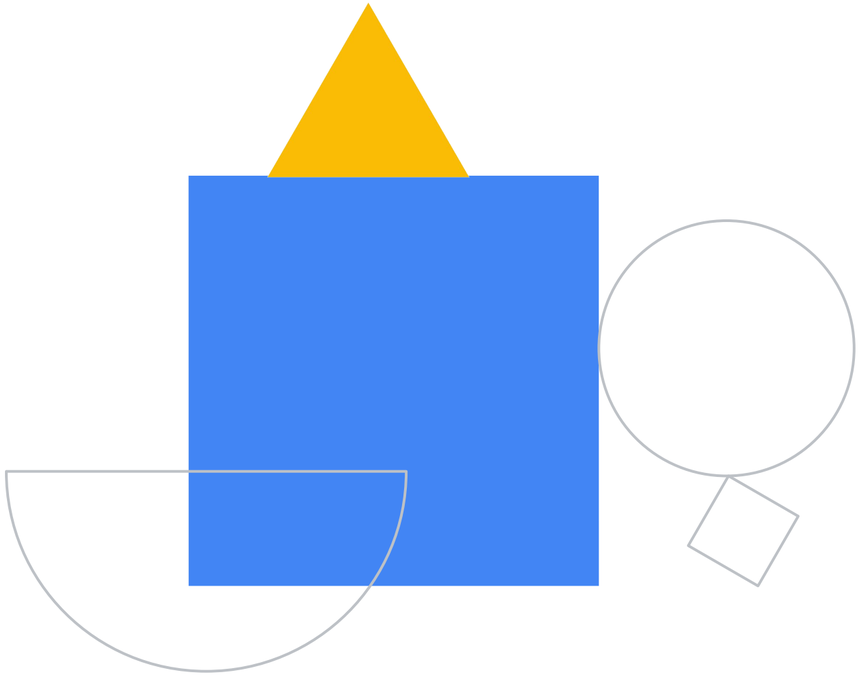
We recommend that you test your uptime check before creating.

TEST | VIEW DETAILS

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Lab Intro

Monitoring and Dashboarding Multiple Projects from a Single Metrics Scope



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Knowledge Check



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Quiz | Question 1

Question

You want to figure out a way to analyze error rate in your load balancing environment. Which interface helps you view a chart with a ratio of 500 responses to all responses?

A. Metrics Explorer

B. MQL

C. Uptime check

D. Liveness probe

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Quiz | Question 1

Answer

You want to figure out a way to analyze error rate in your load balancing environment. Which interface helps you view a chart with a ratio of 500 responses to all responses

A. Metrics Explorer

B. MQL



C. Uptime check

D. Liveness probe

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Quiz | Question 2

Question

You want to be notified if your application is down. What Google Cloud tool makes this easy?

A. Uptime check

B. Health check

C. Readiness probe

D. Liveness probe

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Quiz | Question 2

Answer

You want to be notified if your application is down. What Google Cloud tool makes this easy?

A. Uptime check 

B. Health check

C. Readiness probe

D. Liveness probe

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Quiz | Question 3

Question

What is the name of the project that hosts a metrics scope?

A. Hosting project

B. Monitored project

C. Evaluating project

D. Scoping project

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Quiz | Question 3

Answer

What is the name of the project that hosts a metrics scope?

A. Hosting project

B. Monitored project

C. Evaluating project

D. Scoping project 

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Recap

01 Use Cloud Monitoring to view metrics for multiple cloud projects

02 Explain the different types of dashboards and charts that can be built

03 Create an uptime check

04 Explain and demonstrate the purpose of using MQL for monitoring



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